

Multivariate environmental exposures are reflected in whole-brain functional connectivity and cognition in youth

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ABSTRACT

Each individual's complex, multidimensional environment, known as their "exposome", plays an essential role in shaping cognitive neurodevelopment. Understanding the mechanisms whereby children's exposome influences their development is crucial to facilitate the design of interventions to foster positive developmental trajectories for all youth. Recent work has identified a general exposome factor associated with socio-economic inequality that is strongly related to cognition and individual differences in the spatial organization of functional brain networks in youth. Building on these findings, the current study explores whether alterations in functional connectivity may represent a potential mechanism linking variation in the exposome to cognitive performance. We apply a data-driven, cross-validated, whole-brain machine learning approach, connectome-based statistical inference, to identify patterns of functional connectivity associated with exposome scores among early adolescents enrolled in the Adolescent Brain Cognitive Development (ABCD) Study using data collected during three cognitive tasks and during rest. Additionally, we investigate whether the identified patterns of functional connectivity relate to individual differences in cognitive performance across three domains: General Cognition, Executive Functioning, and Learning/Memory. Models incorporating 10-fold cross-validation over 100 iterations identified consistent functional connections associated with the exposome across task and rest conditions (model performance: $ns = 6137 - 8391$, $rs = 0.34 - 0.44$, $ps < .001$). Results were robust across data collection sites and functional connections common across all significant models were associated with cognitive performance across domains ($ps < 0.0009$). Collectively, these findings reveal that multidimensional environmental exposures are reflected in patterns of functional connectivity and relate to cognitive functioning among youth.

1. Introduction

Each individual's unique environment and lived experiences play an essential role in shaping cognitive neurodevelopment. Previous work

has identified numerous environmental factors that relate to neurodevelopment and cognition, including physical and chemical exposures (e.g. lead poisoning (Marshall et al., 2021) or air pollution (Wodtke et al., 2022; Lubczynska et al., 2021)), psychosocial experiences (e.g.

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caregiver relationship (Davis et al., 2019; Luby et al., 2019) or adverse life events (Cohodes et al., 2021)), and external environments (e.g. socioeconomic resources (Taylor et al., 2020) or cultural influences (Meca et al., 2023)). Most studies to date have examined associations with just one or a few environmental factors at a time. However, it is increasingly accepted that these features often co-occur non-randomly and likely have additive and interactive effects on the developing brain. For example, living in a high poverty neighborhood is associated with greater risk of exposure to air pollutants and traumatic experiences, which in turn are associated with individual differences in cognitive, psychiatric, and neurodevelopmental outcomes (Wodtke et al., 2022; Merrick et al., 2018; McLaughlin and Sheridan, 2016). Consequently, there are increasing efforts to more comprehensively measure each individual's complex, multidimensional environment, referred to collectively as an individual's "exposome". Efforts to characterize the exposome have begun to uncover important associations between large sets of co-occurring environmental features and individual differences in neurodevelopmental outcomes, with potential to ultimately inform policies and interventions that support healthy neurocognitive development (Robinson et al., 2026).

The concept of the "exposome" was first introduced as a complement to the genome in epidemiological studies (Wild, 2005), emphasizing the importance of capturing the totality of co-occurring environmental exposures in addition to genetic influences on physical health. The exposome approach builds upon a long history of prior research examining how specific aspects of early life environments relate to physical and mental health outcomes, including efforts to uncover mechanistic links between adverse experiences and risk for psychiatric illness. Recently, definitions of the exposome have expanded to include psychosocial and sociocultural factors in addition to physical and chemical exposures throughout an individual's lifetime (Vineis and Barouki, 2022). The exposome provides a complementary perspective that improves predictive power by more comprehensively capturing the additive and interactive effects of co-occurring environmental features which are not fully captured by measures of cumulative risk and severity and that may collectively shape health outcomes through complex and interacting pathways. Thus, while previous efforts have made great progress in characterizing mechanistic associations between specific features of the environment and specific neurocognitive outcomes, the broader exposome approach can provide complementary information about how the collective effects of one's multidimensional environment may shape cognitive neurodevelopment more broadly (Vineis and Barouki, 2022).

Emerging work has already begun to leverage large-scale datasets, such as the Adolescent Brain Cognitive DevelopmentSM Study (ABCD Study®) (Volkow et al., 2018), to investigate associations between the exposome and the spatial organization of functional brain networks (Keller et al., 2024), cognitive outcomes (Keller et al., 2024; Brieant et al., 2023; Xiao et al., 2023), and mental health (Xiao et al., 2023; Moore et al., 2022). For example, a multidimensional exposome score reflecting lower socioeconomic status (SES) is associated with poorer cognitive performance and worse mental and physical health (Xiao et al., 2023). In addition, including multidimensional environmental exposures improves brain-based predictions of cognitive performance (Zhi et al., 2024) in held-out data beyond what can be gained by including single environmental features such as household income or neighborhood deprivation indices alone. Prior studies have leveraged bifactor analysis to quantify both a single general exposome factor and several sub-factors that capture specific dimensions of the environment (Keller et al., 2024; Moore et al., 2022). The general exposome factor uncovered using this bifactor approach in the ABCD Study primarily captures co-occurring environmental features related to socio-economic inequality and is strongly associated with symptoms of psychopathology (Moore et al., 2022), cognitive performance (Keller et al., 2024), and individual differences in the spatial organization of functional brain networks (Keller et al., 2024) in youth.

While previous work has revealed that the exposome is reflected in

the static spatial organization of functional brain networks (e.g. functional topography) (Keller et al., 2024), it remains unknown whether the exposome is also reflected in the dynamic activity of functional brain networks during rest or cognitive task performance. Thus, the current study aims to investigate whether the exposome is reflected in patterns of functional connectivity across networks that support cognitive functioning. Coordinated patterns of activity across distributed networks of functionally related brain regions have been shown to give rise to complex cognitive functions (Herbet and Duffau, 2020), and individual differences in whole-brain functional connectivity profiles have been associated with cognitive performance across multiple domains, including fluid intelligence (Finn et al., 2015; Greene et al., 2018), attention (Rosenberg et al., 2016; Yoo et al., 2018), language (Adkinson et al., 2024) and executive functioning (Adkinson et al., 2024). Moreover, developmental changes in functional connectivity profiles (Vasa et al., 2020; Dosenbach et al., 2010; Fair et al., 2007; Luo et al., 2024) are associated with age-related improvements in cognitive performance (Constantinidis and Luna, 2019; Larsen and Luna, 2018; Paz-Alonso et al.; Mitchell et al., 2025), suggesting that functional connectivity profiles may be a promising, developmentally sensitive biomarker of individual differences in youth cognition. Recent work has also shown that functional connectivity relates to environmental exposures, including household socioeconomic status (Rakesh et al., 2021a; Yang et al., 2025), neighborhood disadvantage (Rakesh et al., 2021a), interpersonal unpredictability (Yang et al., 2025), and air pollution (Kusters et al., 2025), with important implications for cognition and mental health (Yang et al., 2025; Rakesh et al., 2021b, 2025). However, prior work has primarily focused on examining only a select subset of environmental features at a time, and the relationship between the full multidimensional exposome and whole-brain functional connectivity patterns during development remains unknown. Characterization of the effects of the exposome as a whole is therefore necessary to delineate the real-world complexity of these relationships.

Here, we apply a modified version of a well-validated, data-driven, whole-brain, machine learning approach, connectome-based predictive modeling (CPM), which we here refer to as connectome-based statistical inference (details below), to identify patterns of functional connectivity associated with exposome scores among early adolescents enrolled in the ABCD Study. Considering recent evidence that different task conditions, i.e. "brain states", are optimal for revealing distinct brain-phenotype relationships with CPM and other predictive modeling approaches (Greene et al., 2018, 2023; Rosenberg and Finn, 2022; Finn, 2021; Lichenstein et al., 2021), we independently examine patterns of functional connectivity using functional magnetic resonance imaging (fMRI) data collected during three different cognitive tasks and during rest. Motivated by evidence that individual differences in cognition during youth are associated with myriad physical and mental health outcomes in adulthood (Cortes Pascual et al., 2019; Klassen et al., 2004; Agha et al., 2019; Richards et al., 2004; Moffitt et al., 2011; Shanmugan et al., 2016; Shamosh et al., 2008; Gow et al., 2011; Hart et al., 2004; Wraw et al., 2015; Batty et al., 2007), we also investigate whether the identified patterns of functional connectivity associated with each individual's exposome also relate to individual differences in cognitive performance across three domains: General Cognition, Executive Functioning, and Learning/Memory (Thompson et al., 2019). Collectively, this work aims to elucidate how multivariate childhood exposures relate to whole-brain functional connectivity and cognition at the transition from childhood to adolescence, a critical developmental period marked by heightened neurobiological sensitivity to environmental influences in brain regions supporting cognitive maturation (Larsen and Luna, 2018).

2. Methods

2.1. Participants

Data were drawn from the Adolescent Brain Cognitive Development

(ABCD) Study ($n = 11,878$), a large-scale study of adolescent neurodevelopment with youth recruited from 21 sites across the United States at age 9–10 and followed prospectively across adolescence (Volkow et al., 2018). The study includes comprehensive demographic, cognitive, and clinical assessments, as well as neuroimaging, including fMRI during 3 tasks and rest (see 2.2.2., below). Details on the recruitment strategy have been previously described (Garavan et al., 2018). All study procedures were approved by a central Institutional Review Board (IRB) at the University of California, San Diego, and several sites also obtained local IRB approval. A parent or guardian provided written informed consent and assent to participate was obtained from participants.

2.2. Measures

2.2.1. Exposome

We leveraged a set of previously derived exposome factors, including a general (overall) factor and 6 subfactor scores, defined using longitudinal bifactor analysis as previously described (Keller et al., 2024). Briefly, general and specific exposome factors were derived from a set of 354 variables comprising youth-report, caregiver-report, and geocoded data covering a wide range of environmental features, which were reduced to a set of 32 data-driven summary scores prior to exploratory structural equation modeling with bifactor rotation (for details see Keller et al., 2024; Volkow et al., 2018). To ensure consistency with prior work, we used the exact same scores as in our previously published study rather than recalculate exposome scores within the present sub-samples of ABCD data across each cross-validation fold, noting that the derivation of the exposome factors was agnostic to measures of neural or cognitive function. Measures of socioeconomic status (SES) (e. g., household income, parent education) show the strongest loadings for the general exposome factor, whereas the 6 subfactors capture more specific dimensions of a child's environment, each of which is orthogonal to the general exposome factor and all other subfactors. These subfactors include School (i.e., school involvement, enjoyment, and performance), Family Values (i.e., Mexican American Cultural Values Scale subscales measuring family values, centrality, and culture), Family Turmoil (i.e., family conflict based on both youth and parent reports), Dense Urban Poverty (i.e., geocoded and parent-report data on youth's neighborhood, including poverty, density, safety, and pollution), Extracurriculars (i.e., involvement in sports and other activities; traumatic brain injury), and Screen Time. The longitudinal bifactor analysis approach ensures that the structure of the exposome scores (i.e. factor loadings) remain consistent across the baseline, year 1 follow-up, and year 2 follow-up timepoints of the longitudinal ABCD Study, allowing future work to examine longitudinal associations. While the structure of the exposome factors remains consistent across timepoints, the values for each exposome score for each participant are still unique to each independent timepoint, allowing us to examine cross-sectional associations between the exposome and functional connectivity at the baseline assessment in the present study.

2.2.2. Neuroimaging data

ABCD Study participants completed a neuroimaging assessment at baseline, including fMRI scanning using 3 T scanners during rest and 3 cognitive tasks: the Stop-Signal Task (SST; response inhibition), the Monetary Incentive Delay Task (MID; reward processing), and the Emotional N-Back Task (EN-back; working memory, affective processing). Details regarding acquisition parameters and task design have been previously described (Casey et al., 2018). Raw time series for the baseline fMRI data were obtained via ABCD's Fast Track Release of raw fMRI data and preprocessed using the Yale Magnetic Resonance Research Center (MRRC) functional connectivity pipeline to generate functional connectivity matrices for each individual for each task and resting-state scan.

Preprocessing was conducted using Bioimage Suite (Joshi et al., 2011) and SPM12 including the entire raw time series, consistent with

prior CPM work (Lichenstein et al., 2021, 2025; Yip et al., 2023; Lichenstein et al., 2023; Yip et al., 2019), and included brain extraction (Lutkenhoff et al., 2014), nonlinear registration to MNI space, and motion correction. The following covariates were regressed out of the data: linear, quadratic, and cubic drifts, mean global signal, mean cerebral-spinal-fluid, mean white-matter signal, and a 24-parameter motion model including six rigid-body motion parameters, six temporal derivatives, and these terms squared. Participants with mean framewise displacement (FD) > 0.2 mm across runs of a given task were excluded from the analysis (Garrison et al., 2023) and variance attributable to mean FD was also accounted for in our primary analyses (see 2.3.1., below; additional post-hoc sensitivity analyses examining effects of head motion can be found in the Supplement). Following preprocessing and exclusion for motion exceeding the 0.2 mm mean FD threshold, there were $n = 8730$ participants with functional connectivity and exposome data that were included across our primary analyses (see Table 1 for demographics; Rest: $n = 9956$ with successfully preprocessed functional connectivity data, i.e., good quality skull stripping and registration, $n = 8877$ following exclusion for head motion > 0.2 mm mean FD, $n = 8391$ with usable functional connectivity, exposome, sex, and age data for inclusion in analysis; MID: $n = 8924$ after preprocessing, $n = 6936$ after motion exclusion, $n = 6672$ with complete data for analysis; SST: $n = 8827$ after preprocessing, $n = 6592$ after motion exclusion, $n = 6385$ with complete data for analysis; EN-back: $n = 8755$ after preprocessing, $n = 6334$ after motion exclusion, $n = 6137$ with complete data for analysis).

Functional data were parcellated into 268 nodes using the Shen 268 atlas (Shen et al., 2013), which includes cortical and subcortical regions, as well as the cerebellum. Functional connectivity between each node pair was then computed using Pearson correlations and correlation coefficients were transformed using the Fisher's r -to- z transformation to yield 268×268 connectivity matrices, termed 'connectomes'. These connectomes represent each participant's whole-brain functional connectivity during each task and rest, which served as the primary input for our primary analyses (described in 2.3.1., below).

2.2.3. Cognition

Cognitive performance was assessed using 3 principal components that have been previously derived from the ABCD Study neurocognitive battery using Bayesian Probabilistic Principal Components Analysis (Thompson et al., 2019), including components reflecting General Cognition, Executive Functioning, and Learning/Memory. These components were computed from seven measures from the NIH Toolbox, as well as the Rey Auditory Verbal Learning Test and the Little Man Task. Scores were obtained via the ABCD Data Exploration and Analysis Portal

Table 1
Demographic characteristics of participants included in primary analyses ($n = 8730$).

Variable	<i>M</i>	<i>SD</i>
Age (months)	119.20	7.52
Sex	N	%
Female	4431	50.76
Male	4299	49.24
Household Income	N	%
< \$50 K	2164	24.79
≥ \$50 K & < \$100 K	2337	26.78
≥ \$100 K	3524	40.38
Race	N	%
White	4817	55.18
Black	1154	13.22
Hispanic	1670	19.13
Asian	175	2.00
Other	913	10.46
Parent Education	N	%
High school or lower	1362	15.60
Undergraduate or Associates	5069	58.06
Advanced degree	2290	26.23

(DEAP).

2.3. Analysis

2.3.1. Exposome network identification

Here, we adapted a cross-validated machine learning approach, connectome-based predictive modeling (CPM), to identify neural networks that are statistically associated with individual differences in the exposome (Lichenstein et al., 2021, 2025; Yip et al., 2023; Lichenstein et al., 2023; Yip et al., 2019) (hereafter referred to as connectome-based statistical inference). We use this term to recognize that the exposome was not calculated separately for individual folds, as described above. Nonetheless, this approach was selected to leverage an established analytic framework that incorporates the full multivariate connectome as well as rigorous cross-validation (here, 10-fold cross-validation across 100 iterations to ensure results were not driven by a random split of the data). We also include sensitivity analyses to assess variation in model performance across ABCD data collection sites (see 2.3.2., below). We note that while the exposome is treated as the outcome variable in these analyses, consistent with the CPM-style framework, the current application of connectome-based statistical inference aims to identify patterns of functional connectivity associated with individual differences in the exposome without implying any causality or directionality between brain and environmental variation. Individual participant connectomes computed from MID, SST, EN-back, and resting-state data served as the input for our primary analyses. We conducted 10-fold cross-validation, where 90% of the sample was used as training data and 10% was used as testing data within each fold. For the initial feature selection step, linear regression was used to identify edges that were significantly related to the outcome at $p < 0.05$ in the training data. To account for effects of sex, age, and head motion, these variables were regressed out of the exposome scores within each fold and the residuals of this analysis were used as the primary outcome measure. Edges for which stronger connectivity was a positive predictor of the residualized exposome scores and those for which reduced connectivity was a positive predictor of the residualized exposome scores were summed separately to generate positive and negative summary scores, respectively.

This was followed by a model generation step, in which a linear model was fit to predict the residualized exposome scores from the positive and negative summary scores. This model was then applied to the testing data to generate out-of-sample predictions of residualized exposome scores. The covariates in the test set were entered into the covariate regression model generated from the training set to yield predicted values reflecting the contribution of confounding variables. The final predicted exposome score was computed as the sum of the predicted residualized exposome score and the predicted covariate effects. Finally, model performance was evaluated based on the Pearson correlation between predicted and actual exposome scores across all participants in the test set.

To ensure that the model was not overfit to one 10-fold split, we repeated the analysis described above 100 times, using random splits of the data on each iteration, and calculated mean performance across all iterations. Because analyses across folds are not entirely independent, we tested for statistical significance via permutation testing, where the correspondence between connectomes and exposome scores was randomly shuffled and the model was rerun 1000 times to create a null distribution. Models were run separately for each task and rest, as well as separately for the general exposome factor and the 6 subfactors. To account for multiple comparisons across four brain states and seven exposome factors, a Bonferroni-corrected threshold of $p < 0.002$ was used to evaluate model significance. We then visualized the anatomy of identified networks based on overlap with canonical neural networks (Noble et al., 2017) and anatomical location of high-degree nodes (i.e., nodes with the greatest number of significant edges), as in prior CPM work (Lichenstein et al., 2021, 2025; Yip et al., 2023; Lichenstein et al., 2023; Yip et al., 2019). Cosine similarity was used to quantify similarity

between networks identified across task conditions.

2.3.2. Sensitivity analyses

To assess variation in model performance across ABCD data collection sites, follow-up analyses were conducted quantifying the association between predicted and actual general exposome scores in each brain state across all participants from each site independently. To determine whether model performance varied systematically across data collection sites based on site differences in exposome variability, we then compared these association values with the standard deviation of exposome scores at each site. We further evaluated variation in model performance based on head motion by quantifying the association between predicted and actual general exposome scores among participants with mean FD < 0.15 mm and those with mean FD ≥ 0.15 mm (see Supplement).

We conducted an additional sensitivity analysis to determine whether our interpretations of which large-scale functional brain networks comprise the positive exposome network would differ when comparing this network to individualized functional neuroanatomy rather than a standard group-averaged atlas of functional brain regions. To do so, we leveraged $n = 6972$ person-specific functional atlases that we had previously defined in the ABCD Study dataset in our prior work (Keller et al., 2023a) using spatially-regularized non-negative matrix factorization. These atlases represent the functional topography of 17 personalized functional networks (PFNs) that vary in size, shape, and spatial location across individuals. We then identified the top 10% of highest-degree Shen atlas nodes in the positive exposome network from the present study, converted these nodes to a binarized vertex-wise map in Freesurfer fsLR (59k) space and quantified which of the 17 PFNs had the highest network loading at each vertex of the positive exposome network for each individual participant using a winner-take-all approach. We also repeated this analysis using a group-average hard parcellation of the same 17 PFNs derived from an independent dataset (the Philadelphia Neurodevelopmental Cohort) as described in previous work (Keller et al., 2023a) to further quantify differences in vertex-wise overlap between person-specific parcellations and a group-average parcellation.

2.3.3. Associations with cognition

To investigate whether the patterns of functional connectivity associated with exposome scores also relate to individual differences in cognition, we performed linear regression analyses to quantify the relationship between network strength of a consensus network (i.e., the network corresponding to edges consistently identified across brain states, details in Results) and each of the three cognitive scores (i.e., General Cognition, Executive Function, Learning/Memory), controlling for age and sex (head motion was not included as a covariate because it was accounted for during network identification). Significance was assessed using permutation testing, where the correspondence between network strength and each cognitive domain was shuffled 10,000 times to create a null distribution.

3. Results

3.1. Exposome network identification

3.1.1. Model performance

Models incorporating data collected during all tasks and rest were successful in predicting the general exposome factor ($r_s = 0.34$ – 0.44 , $p_s < .001$) and each of the 6 subfactors ($r_s = 0.08$ – 0.25 , $p_s < .001$; see Table 2 for individual model performance statistics) in held-out participants.

3.1.2. Network anatomy

Consistent with prior CPM work, identified networks are complex and include edges connecting regions throughout the brain. Notably,

Table 2
Model Performance.

Brain State	Network	General Exposome			School			Family Values			Family Turmoil			Dense Urban Poverty			Extracurriculars			Screen Time		
		R	R ² (%)	P	R	R ² (%)	P	R	R ² (%)	P	R	R ² (%)	P	R	R ² (%)	P	R	R ² (%)	P	R	R ² (%)	P
Rest	Positive	0.39	15.56	< 0.001	0.18	3.12	< 0.001	0.09	0.82	< 0.001	0.09	0.88	< 0.001	0.25	6.36	< 0.001	0.10	0.97	< 0.001	0.17	2.88	< 0.001
Rest	Negative	0.45	20.60	< 0.001	0.18	3.09	< 0.001	0.08	0.71	< 0.001	0.08	0.71	< 0.001	0.23	5.46	< 0.001	0.10	1.08	< 0.001	0.16	2.66	< 0.001
Rest	Combined	0.44	18.96	< 0.001	0.18	3.12	< 0.001	0.09	0.82	< 0.001	0.09	0.87	< 0.001	0.24	6.00	< 0.001	0.10	1.10	< 0.001	0.17	2.84	< 0.001
SST	Positive	0.32	10.24	< 0.001	0.16	2.70	< 0.001	0.08	0.61	< 0.001	0.09	0.86	< 0.001	0.23	5.18	< 0.001	0.08	0.71	< 0.001	0.12	1.55	< 0.001
SST	Negative	0.35	12.51	< 0.001	0.16	2.59	< 0.001	0.07	0.44	< 0.001	0.10	1.01	< 0.001	0.22	4.78	< 0.001	0.09	0.84	< 0.001	0.13	1.70	< 0.001
SST	Combined	0.34	11.81	< 0.001	0.16	2.67	< 0.001	0.08	0.57	< 0.001	0.10	0.97	< 0.001	0.23	5.08	< 0.001	0.09	0.83	< 0.001	0.13	1.68	< 0.001
MID	Positive	0.32	10.08	< 0.001	0.15	2.23	< 0.001	0.09	0.84	< 0.001	0.08	0.62	< 0.001	0.23	5.45	< 0.001	0.10	1.07	< 0.001	0.12	1.39	< 0.001
MID	Negative	0.36	13.14	< 0.001	0.15	2.21	< 0.001	0.09	0.88	< 0.001	0.08	0.67	< 0.001	0.22	4.65	< 0.001	0.10	1.01	< 0.001	0.12	1.41	< 0.001
MID	Combined	0.35	12.08	< 0.001	0.15	2.21	< 0.001	0.10	0.94	< 0.001	0.08	0.68	< 0.001	0.22	5.06	< 0.001	0.11	1.13	< 0.001	0.12	1.43	< 0.001
EN-back	Positive	0.37	13.65	< 0.001	0.18	3.39	< 0.001	0.10	1.10	< 0.001	0.10	0.98	< 0.001	0.24	5.99	< 0.001	0.10	1.01	< 0.001	0.13	1.73	< 0.001
EN-back	Negative	0.40	16.39	< 0.001	0.18	3.09	< 0.001	0.10	1.05	< 0.001	0.10	1.05	< 0.001	0.22	5.03	< 0.001	0.10	0.99	< 0.001	0.13	1.68	< 0.001
EN-back	Combined	0.40	15.84	< 0.001	0.18	3.28	< 0.001	0.11	1.20	< 0.001	0.10	1.03	< 0.001	0.23	5.52	< 0.001	0.10	1.07	< 0.001	0.13	1.77	< 0.001

Note. R² = 1 - (error SS)/(total SS). Abbreviations: SST: Stop-Signal Task; MID: monetary incentive delay task; EN-back: Emotional N-back Task.

consistent patterns emerged across general exposome networks identified using data from each task and rest (cosine similarity based on canonical functional network edge counts > 0.9, p s < 0.001; see Table 3). Therefore, we focus primarily on a consensus network composed of edges present in all identified networks, including 2211 positive network edges and 1839 negative network edges (see Fig. 1; see Figure S1 for anatomy of general exposome factor networks generated from each task and rest condition separately and Figures S2–S5 for anatomy of 6 subfactor networks generated from each task and rest).

Examining network anatomy based on overlap with canonical neural networks (Fig. 1A), higher general exposome scores, which reflect higher SES, residing in safer and less crowded neighborhoods, and having greater involvement in extracurricular activities, were associated with greater connectivity between the motorsensory network and medial frontal, frontoparietal, salience, and subcortical networks. Higher general exposome scores were also associated with lower connectivity between the frontoparietal network and salience, subcortical, and cerebellar networks, as well as between motorsensory and visual association networks. Examining network anatomy based on nodes with the most significant edges, i.e., high degree nodes (Fig. 1B), the highest degree nodes of the positive network included bilateral putamen, bilateral insula, and bilateral cerebellum. The highest degree nodes of the negative network included bilateral precuneus and bilateral cerebellar regions.

3.2. Sensitivity analyses

To assess whether relationships between the identified networks and exposome scores vary across data collection sites, we conducted post-hoc sensitivity analyses by assessing the correlation between predicted and actual exposome scores within each site individually. Model performance remained significant across all sites for rest, and most sites across tasks (95.2% of sites for EN-back and MID, 85.7% of sites for SST), suggesting that our primary models are robust to site variation. Nonetheless, the strength of the association varied (see Fig. 2). Notably, across all tasks and rest, model performance (i.e., the relationship between predicted and actual general exposome scores) was relatively stronger within the LA Children's Hospital, South Carolina, New York, University of Florida, Maryland, Michigan, Missouri, and Connecticut sites, and relatively weaker within the Colorado, Florida International, Oregon, Minnesota, Utah, and Vermont sites, compared to model performance in the full dataset. This pattern appears to be at least partially attributable to greater variation in exposome scores across participants at the sites characterized by better model performance (see Figure S6).

We also conducted a sensitivity analysis to determine whether the interpretation of which functional brain networks comprise the exposome network differs when compared with individualized functional neuroanatomy. To test this, we compared the top 10% of highest degree nodes in the positive exposome network to personalized functional network (PFN) atlases for each participant that were previously derived in this dataset using spatially-regularized non-negative matrix factorization (Keller et al., 2023a). This analysis revealed that the positive exposome network primarily comprised Auditory (PFN 16), Ventral Attention (PFNs 9 and 7), and Default Mode (PFN 1) networks (see Figure S7), highlighting that interpretations of the spatial topography of networks derived from connectome-based statistical inference may differ when evaluated with respect to individually-defined functional

Table 3

Cosine similarity between general exposome networks identified across brain states.

	SST	MID	EN-back
MID	0.953		
EN-back	0.954	0.940	
rest	0.953	0.902	0.935

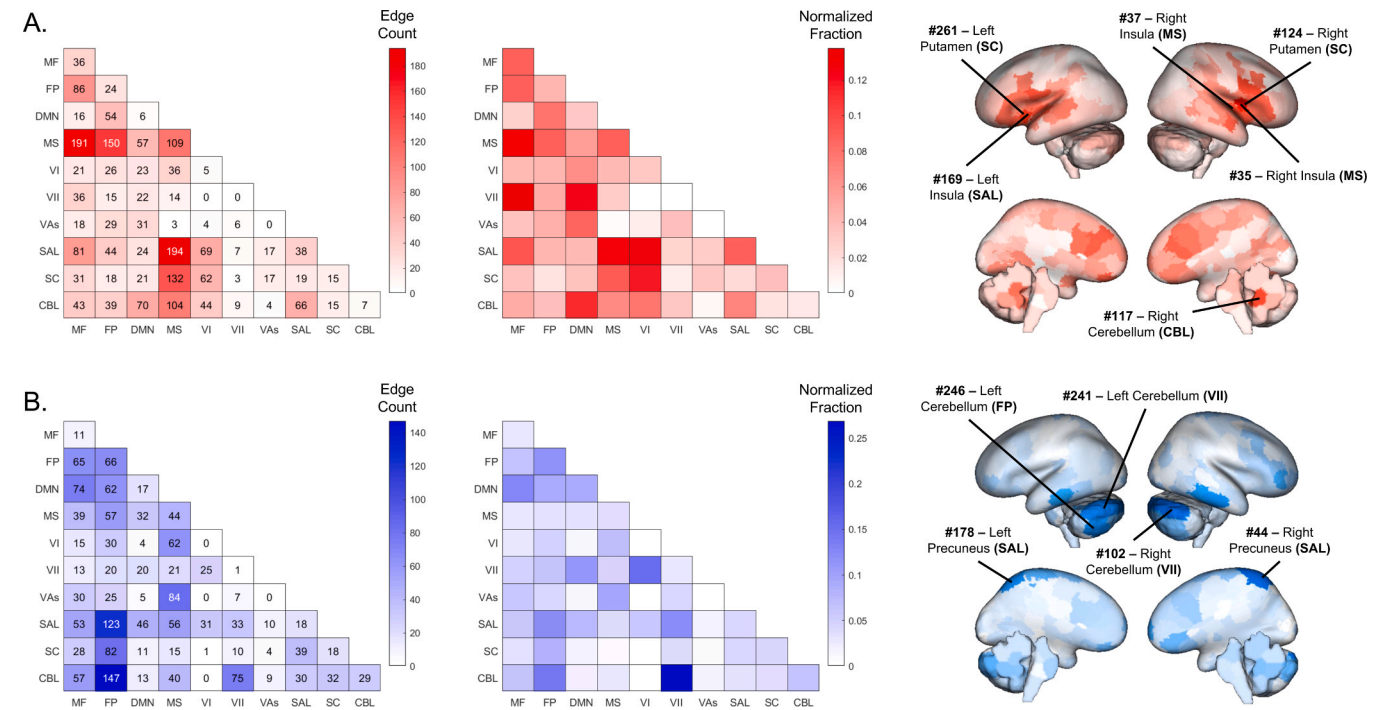


Fig. 1. Consensus General Exposome Network Anatomy. Anatomy of edges identified in models conducted across all brain states (i.e., rest, MID, SST, EN-back) visualized based on overlap with canonical neural networks and nodal degree. Overlap with canonical neural networks is presented based on raw edge count and normalized by network size based on the total number of possible within/between-network edges (i.e., normalized fraction). Panel A shows positive network edges (n = 2211), for which greater connectivity is predictive of higher general exposome scores; Panel B shows negative network edges (n = 1839), for which reduced connectivity is predictive of higher general exposome scores. MF: medial frontal, FP: frontoparietal, DMN: default mode network, MS: motorsensory, VI: visual 1, VII: visual 2, VAs: visual association, SAL: salience, SC: subcortical, CBL: cerebellar.

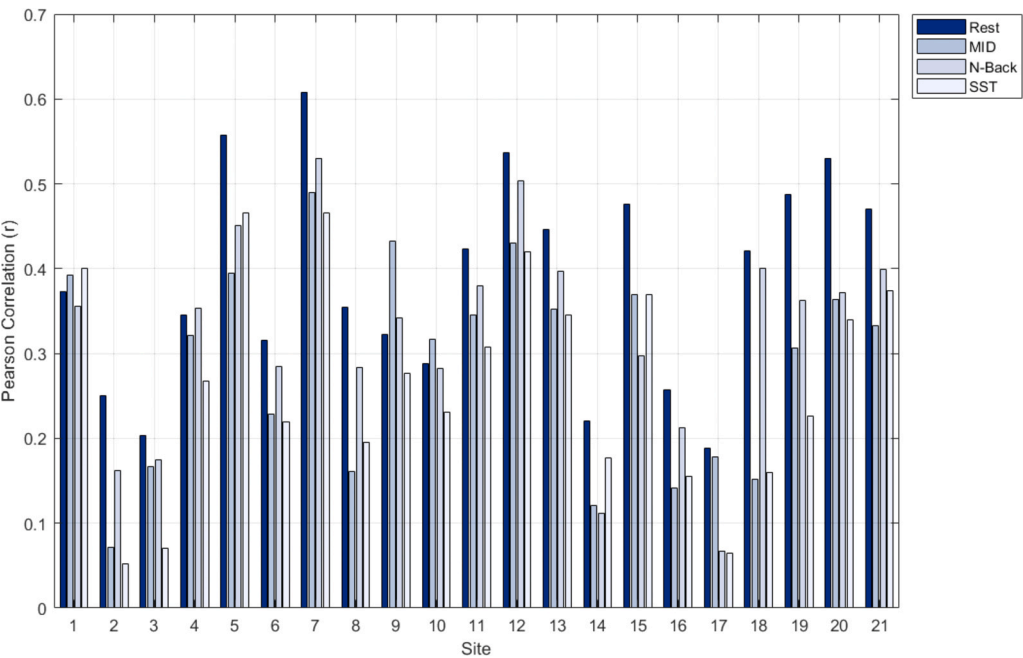


Fig. 2. Variation in General Exposome Model Performance Across Sites. The association between model predicted and observed general exposome scores is presented within each site separately for models run on data collected during rest, MID, EN-back, and SST task performance. Model performance was significant across all sites based on permutation testing.

brain networks rather than standard group-average atlases.

3.3. Associations with cognition

Greater consensus network strength was significantly positively associated with General Cognition (β 's = 0.15–0.17), Executive

Function (β 's = 0.03–0.07), and Learning/Memory (β 's = 0.06–0.09; see Table 4 for individual model statistics and Fig. 3).

4. Discussion

Individuals' early environments vary in myriad ways, with important implications for neurodevelopment. The current study applied connectome-based statistical inference – an adapted version of a well-validated, whole-brain, statistical approach, connectome-based predictive modeling – to identify patterns of functional connectivity predictive of individual differences in a general exposome factor capturing variation in over 350 variables reflecting different aspects of the environment. Consistent neural features were identified across models using data from three different cognitive tasks and rest. Furthermore, this general exposome network was robust to data collection site and predictive of cognitive performance across three domains: General Cognition, Executive Functioning, and Learning/Memory. Collectively, these findings provide novel insight into how multidimensional environmental exposures are reflected in patterns of functional connectivity and relate to cognitive functioning among youth.

Overall, early environments characterized by higher socioeconomic status were associated with greater connectivity among motor/sensory, frontoparietal, medial frontal, salience, and subcortical network regions, coupled with reduced connectivity between frontoparietal, salience, cerebellar, and visual network regions. This pattern of results is consistent with prior ABCD studies demonstrating altered motor/sensory (Zhi et al., 2024; Acosta-Rodriguez et al., 2025; Michael et al., 2024), frontoparietal (Acosta-Rodriguez et al., 2025), salience (Acosta-Rodriguez et al., 2025), cerebellar (Zhi et al., 2024), and subcortical (Zhi et al., 2024; Michael et al., 2024) connectivity in association with socioeconomic resources (Michael et al., 2024), neighborhood deprivation (Zhi et al., 2024; Acosta-Rodriguez et al., 2025) and differences in the family environment (Zhi et al., 2024) (i.e., parent income and education), as well as findings from other studies reporting associations between air pollution and connectivity of motor/sensory and salience networks (Kusters et al., 2025). These results also align with findings from prior ABCD work applying predictive modelling to elucidate functional connectivity-based substrates of cognitive performance, including greater motor/sensory connectivity with medial frontal and subcortical regions and lower connectivity between frontoparietal and salience networks (Chen et al., 2022). The central role for frontoparietal, motor/sensory, visual and subcortical network connectivity is also consistent with prior work in other datasets applying CPM to predict various domains of cognitive performance (Greene et al., 2018; Adkinson et al., 2024; Lv et al., 2024).

The current pattern of results also aligns with prior work suggesting that hierarchical development along a sensorimotor-association (S-A) axis may support age-related improvements in cognitive functioning (Keller et al., 2023b). In particular, greater segregation at each pole of the S-A axis, coupled with greater integration of middle-axis and frontoparietal systems, is thought to facilitate a balance of efficient within-system processing and cross-system communication to support developmental improvements in cognition. In line with this framework,

our general exposome network consists of high within-network connectivity of the motor/sensory and default mode networks and high between-network connectivity of the frontoparietal network, potentially reflecting greater segregation at sensorimotor and association poles of the S-A axis, along with the cross-system communication needed to optimize cognitive performance. This is consistent with recent studies demonstrating that neural effects of environmental exposures vary across the S-A axis (Keller et al., 2024; Michael et al., 2024; Tooley et al., 2024; Zhao et al., 2024). Notably, prior work defining and characterizing the S-A axis has focused predominantly on cortical-cortical connectivity patterns (Luo et al., 2024; Keller et al., 2023b; Sydnor et al., 2023), whereas the current study results align with recent research (Michael et al., 2024) in also highlighting an important role for subcortical and cerebellar network connectivity features. Accordingly, future work is needed to expand upon how early environments may relate to subcortical and cerebellar connectivity, and how connectivity of these networks fits within the S-A axis hierarchy to support cognitive neurodevelopment.

The present findings complement earlier work showing that the exposome is associated with spatial patterns of functional network topography, particularly for frontoparietal, default mode, and attention networks (Keller et al., 2024). Collectively, these two sets of findings suggest convergent relationships between environmental variation and functional network topography and connectivity, particularly for the frontoparietal network. These findings are consistent with prior work suggesting that the transition to adolescence is a key period for frontoparietal network development (Dong et al., 2021), which is thought to play a critical role in age-related improvements in cognitive functioning (Keller et al., 2023b). Notably, prior work showed that the general exposome was only weakly related to individual differences in functional network topography of sensory and motor networks, but the present study reveals prominent associations with the functional connectivity of the motor/sensory network. This divergence is notable in light of research suggesting individual differences in neural topography may confound functional connectivity analyses (Bijsterbosch et al., 2018) and demonstrates that it is feasible to identify both shared and distinct associations of the exposome with functional network topography and connectivity. Together, these two sets of results suggest that the topography and connectivity of the motor/sensory network may differ in developmental timing and/or susceptibility to effects of environmental exposures at the onset of adolescence.

It is interesting to note that patterns of functional connectivity predictive of general exposome scores were remarkably consistent across different brain states, including three cognitive tasks and a rest period. While there is a growing body of literature highlighting the importance of brain state for optimizing brain-behavior models (Greene et al., 2018, 2023; Rosenberg and Finn, 2022; Finn, 2021; Finn et al., 2017), less is known about the importance of brain state for optimizing brain-environment models that seek to characterize functional connectivity correlates of early-life environments. The current work, which leveraged an aggregate measure comprising a wide range of multivariate environmental factors, suggests that co-occurring features of childhood environments may have far-reaching impacts on neurodevelopment that are evident across domains and independent of current brain state. Future studies may seek to further disentangle the additive and interactive effects of specific physical/chemical, psychosocial, or socioeconomic environmental factors on general and brain-state-specific functional connectivity patterns.

The current study leverages the large sample size and rich phenotyping of the ABCD Study dataset and applies a robust connectome-based approach incorporating stringent cross-validation to elucidate patterns of functional connectivity associated with multivariate environmental exposures and youth cognition. Nonetheless, our results should be interpreted in light of several important limitations. While our models were conducted with 10-fold internal cross-validation across 100 iterations, we do not have an external validation sample. Therefore,

Table 4
Associations between General Exposome Combined Network Strength and Cognition across Brain States.

Brain State	General Cognition		Executive Functioning		Learning/Memory	
	β	Permuted Sig.	β	Permuted Sig.	β	Permuted Sig.
Rest	0.17	< 0.0001	0.06	< 0.0001	0.08	< 0.0001
EN-back	0.15	< 0.0001	0.04	< 0.0001	0.07	< 0.0001
SST	0.15	< 0.0001	0.03	0.0008	0.06	< 0.0001
MID	0.16	< 0.0001	0.07	< 0.0001	0.09	< 0.0001

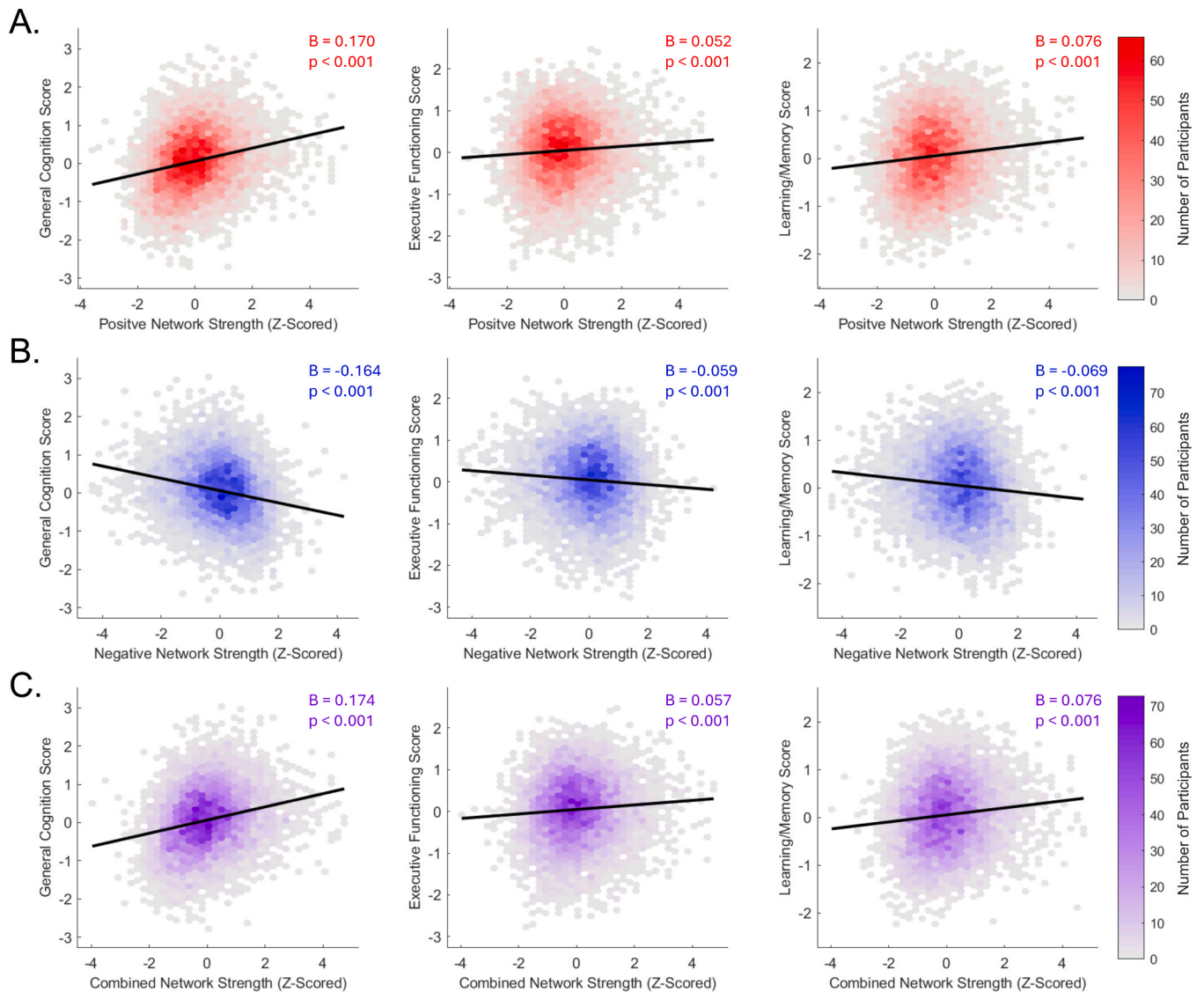


Fig. 3. Associations Between Consensus General Exposome Network Strength and Cognition. The positive (A), negative (B), and combined (C) consensus general exposome network is significantly associated with General Cognition, Executive Function, and Learning/Memory during rest.

future research is needed to externally replicate our functional connectivity findings and to assess whether the exposome network we identified is associated with cognitive functioning in independent datasets. Relatedly, ABCD Study participants with high-quality neuroimaging data are known to differ in their sociodemographic characteristics from the general population, including higher socioeconomic status among participants with low-motion data, such that the subgroup of ABCD Study participants included in neuroimaging studies may not accurately represent the general population (Cosgrove et al., 2022). Accordingly, it is particularly important for future efforts to assess whether the current findings generalize to samples whose sociodemographic makeup is representative of the population. Furthermore, the exposome was computed across all participants resulting in a lack of total independence between training and testing samples. Nonetheless, the aim of the current work was to identify patterns of connectivity associated with variation in the exposome, not to achieve true statistical prediction. Accordingly, we utilized a cross-validated CPM-style analysis approach to increase statistical rigor and adopt the term ‘connectome-based statistical inference’ to acknowledge that the current analyses do not fully align with the full predictive modeling CPM framework. Additionally, the current analyses aimed to investigate cross-sectional associations among the exposome, functional

connectivity, and cognition to determine whether environmental exposures are reflected in patterns of functional coherence. Future work may examine whether the network we identified also relates to cognition at later timepoints, as well as how this network evolves across development in the context of both stable and changing environments. Moreover, given the observational design of the ABCD Study, we cannot infer causation between the exposome and identified patterns of functional connectivity. Future research may also leverage longitudinal mediation models to more directly investigate causal relationships among environmental exposures, functional connectivity, and behavior. Given that the exposome has previously been shown to be associated with spatial patterns of functional topography (Keller et al., 2024) and that leveraging individual-specific functional brain atlases led us to different interpretations of which large-scale brain networks comprise the positive exposome network, future work may continue to leverage precision functional mapping in tandem with CPM approaches (pCPM) to better describe CPM-derived networks with respect to each individual’s unique functional neuroanatomy. Furthermore, the general exposome does not account for all aspects of the environment that have previously been shown to be associated with brain function (e.g., racial discrimination (Elbasheir et al., 2024; Muscatell et al., 2022) and traumatic events (Zhu et al., 2022); Hardi et al., 2025) and our study did not

investigate non-environmental (i.e., genetic) influences on functional connectivity and cognition. Therefore, future work is needed to incorporate more facets of children's early environments and to examine how genetic and environmental factors interact to shape cognitive neurodevelopment.

Finally, it is worth noting that the holistic "exposome" approach used here is considered complementary to more fine-grained mechanistic investigations of the effects of specific environmental exposures. While some studies aim to probe or causally manipulate a single environmental exposure at a time in a controlled manner to maximize *mechanistic* insight, the exposome approach employed here aims to comprehensively capture a large multitude of diverse, co-occurring, and interrelated exposures to maximize *predictive* power (e.g. predictions of future health outcomes, or, in our case, predictions in held-out samples). Here, the goal of predicting is to maximize statistical rigor by ensuring that the findings we identify are able to generalize to data previously unseen by the model. This exposome approach comes with important tradeoffs: it allows us to model environmental features as they occur more naturally, as single environmental exposures rarely occur in isolation and instead tend to non-randomly co-occur with other types of exposures, and often yields stronger, more generalizable associations with neurobiological features. Yet, this exposome approach is limited in its ability to yield mechanistic insights about subtler effects that specific exposures may have on more granular neurobiological features (e.g. an association between lead poisoning and functional connectivity in a particular brain region). Future mechanistic studies may build upon the results presented here to identify the effects of specific environmental exposures on functional connectivity features and identify those that may serve as the most impactful targets for the development of novel interventions to support healthy cognitive neurodevelopment.

Collectively, the current results show that multidimensional features of children's early environments are reflected in whole-brain patterns of functional connectivity at the onset of adolescence and relate to cognitive functioning across multiple key domains. These findings add to a growing literature elucidating how the environment impacts neurodevelopment and cognition among youth (Zhou et al., 2025; Qiu et al., 2025). Our observation that common aspects of functional connectivity are associated with the exposome across multiple key brain states (i.e., during rest, inhibitory control, reward processing, and emotional working memory task performance) highlights the far-reaching impact of the environment, as well as the potential impact of translational interventions geared toward improving children's environments for fostering positive developmental trajectories (Uddin, 2025). Indeed, recent ABCD findings have demonstrated that associations between poverty and altered brain structure are mitigated in states that provide more generous support for low-income families (Weissman et al., 2023), highlighting the potential for policy changes to improve youth neurodevelopment. Future work is needed to investigate how the observed patterns of functional connectivity evolve across development and relate to longer-term trajectories of wellbeing.

CRedit authorship contribution statement

Menessa Metayer: Writing – review & editing, Visualization. **Joliza Maynard:** Writing – review & editing, Visualization. **Corey Horien:** Writing – review & editing, Resources, Data curation. **Suhani Suneja:** Writing – review & editing, Visualization. **Sarah W. Yip:** Writing – review & editing, Conceptualization. **Heather Robinson:** Writing – review & editing, Writing – original draft. **Ran Barzilay:** Writing – review & editing, Resources. **Yihe Weng:** Writing – review & editing, Formal analysis, Data curation. **Amar Ojha:** Formal analysis, Writing – review & editing. **Marzieh Babaeianjelodar:** Writing – review & editing, Data curation. **Arielle S. Keller:** Writing – review & editing, Supervision, Data curation, Project administration, Conceptualization. **Lester Rodriguez:** Writing – review & editing, Visualization, Supervision. **R. Todd Constable:** Writing – review & editing, Resources. **Abigail S.**

Greene: Writing – review & editing, Resources, Data curation. **Sarah D. Lichenstein:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Conceptualization. **Tyler M. Moore:** Writing – review & editing, Resources.

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Declaration of Competing Interest

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ABCD Consortium investigators designed and implemented the study and/or provided data but did not necessarily participate in the analysis or writing of this report. This manuscript reflects the views of the authors and may not reflect the opinions or views of the NIH or ABCD Consortium investigators.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.dcn.2026.101788](https://doi.org/10.1016/j.dcn.2026.101788).

Data availability

Data used in the preparation of this article were obtained from the Adolescent Brain Cognitive DevelopmentSM (ABCD) Study (<https://abcdstudy.org>), held in the NIMH Data Archive (NDA).

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